Chromium-52

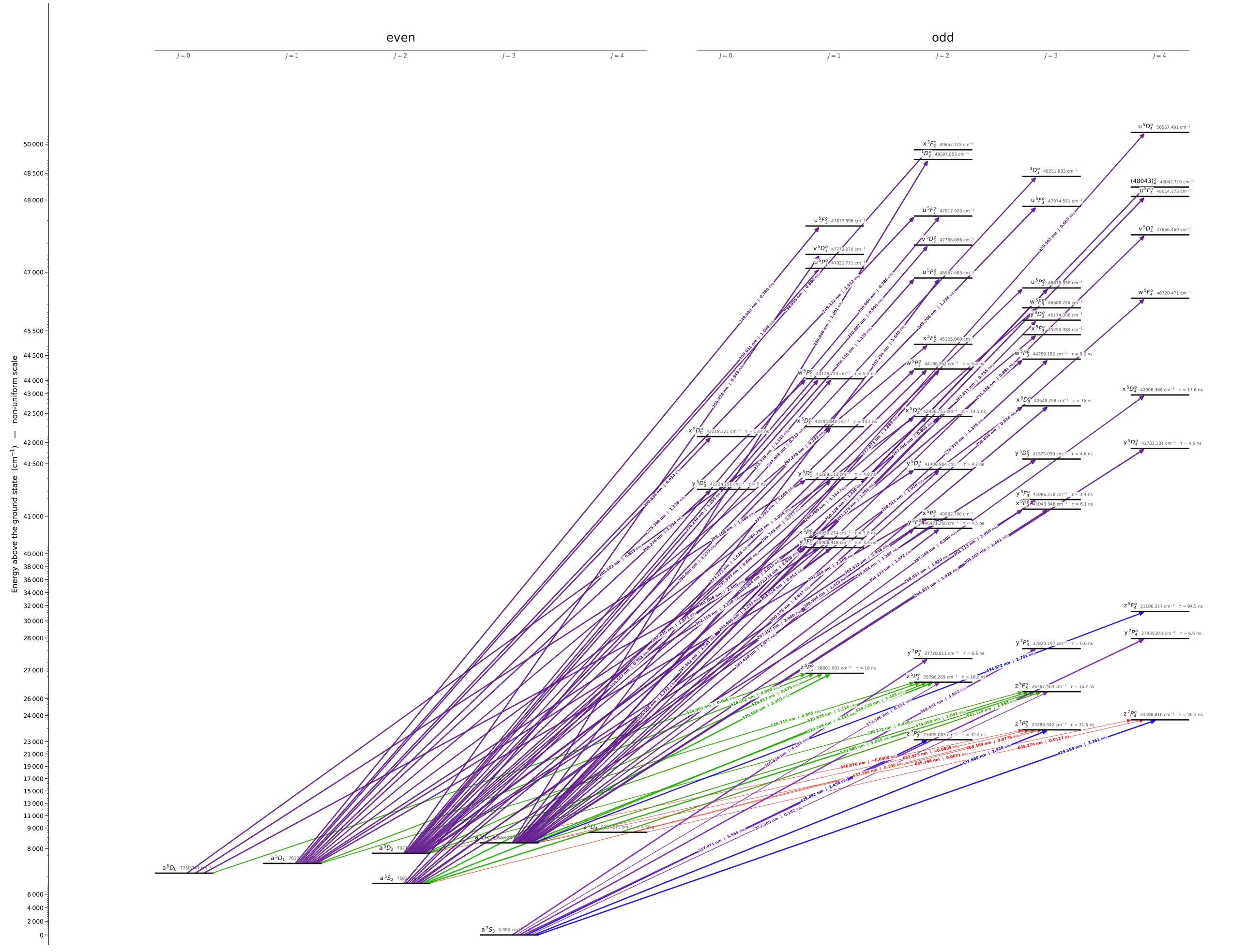
^{52}Cr (Cr I), nuclear spin $I = 0$. Strongest classified transitions below 2 µm (NIST): 56 levels, 87 lines.

Cr^{+} ionisation limit 54575.60 cm^{-1} = 6.76651 eV (183.232 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|\langle J \| er \| J' \rangle|$ in ea_0
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model or semi-empirical calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).

Arrow colour = vacuum wavelength (nm)

300 400 500 600 700 800 1000 1200 1500 2000



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	\|⟨J‖er‖J′⟩\| (ea₀)	A (s⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
a $^7S_3$ – y $^7P_4^o$	357.9708	357.8686	837.4775	5.565	1.520e+08	24.11	100	strong UV resonance line a7S3 - y7P4
a $^7S_3$ – y $^7P_3^o$	359.4511	359.3485	834.0285	4.922	1.510e+08	24.11	99.7	strong UV resonance line a7S3 - y7P3
a $^7S_3$ – y $^7P_2^o$	360.6357	360.5329	831.2888	4.181	1.510e+08	24.11	99.7	strong UV resonance line a7S3 - y7P2
a $^7S_3$ – z $^7P_4^o$	425.5534	425.4336	704.4768	3.361	3.300e+07	5.253	100	main cooling line: Zeeman slower, MOT, imaging
a $^7S_3$ – z $^7P_3^o$	427.6000	427.4797	701.1049	2.926	3.170e+07	5.053	99.9	optical pumping to m_J = -3 and "depumper" towards 5S2 /...
a $^7S_3$ – z $^7P_2^o$	429.0924	428.9717	698.6664	2.458	3.100e+07	4.943	99.8	429.1 nm line
a $^5S_2$ – z $^7P_3^o$	633.1842	633.0091	473.4680	0.160	2.900e+04	5.053	0.0914	633.2 nm repumper 5S2 -> 7P3
a $^5D_2$ – z $^7P_3^o$	646.8765	646.6978	463.4462	≈0.030	9.647e+02	5.053	0.00304	646.9 nm repumper 5D2 -> 7P3
a $^5D_3$ – z $^7P_4^o$	649.1976	649.0182	461.7892	0.007	4.200e+01	5.253	0.000127	intercombination leak of the cooling line into metastable...
a $^5D_3$ – z $^7P_3^o$	653.9727	653.7921	458.4174	≈0.063	4.168e+03	5.053	0.0131	654.0 nm repumper 5D3 -> 7P3
a $^5D_4$ – z $^7P_4^o$	658.2741	658.0923	455.4219	0.013	1.270e+02	5.253	0.000385	intercombination leak of the cooling line into metastable...
a $^5D_4$ – z $^7P_3^o$	663.1842	663.0011	452.0500	0.078	6.000e+03	5.053	0.0189	663.2 nm repumper 5D4 -> 7P3

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
a $^7S_3$ – z $^7P_4^o$	425.553	53-52	10.7(17)
a $^7S_3$ – z $^7P_3^o$	427.600	53-52	18(23)
a $^5S_2$ – z $^7P_3^o$	633.184	53-52	-155(53)
a $^5D_3$ – z $^7P_3^o$	653.973	53-52	670(28)
a $^5D_4$ – z $^7P_3^o$	663.184	53-52	819(28)

Shift = ν(first isotope) − ν(second isotope).